Karanam Rao

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Leveraging Vision Transformers for Scalable Plant Health Monitoring: Early Disease Detection and Precision Agriculture Applications

[1]Pottabathini Sreenidhi
Department of Engineering
Anurag University, Hyderabad, India
21eg112c40@anurag.edu.in

[2]Rao Shreyank Reddy
Department of Engineering
Anurag University,Hyderabad,India
21eg112c42@anurag.edu.in

[3]N Manish Kumar Reddy
Department of Engineering,
Anurag University, Hyderabad,India
21eg112c64@anurag.edu.in

[4]Dr G L Anand Babu
Assistant Professor
School of Engineering,
Anurag University Hyderabad,India
anandbabuit@anurag.edu.in

[5]N Nagalakshmi
Assistant Professor
School of Engineering
Anurag University Hyderabad,India
nagalakshmiit@anurag.edu.in

Abstract - Ensuring plant health is vital for sustainable agriculture, and advancements in deep learning have enabled automated monitoring techniques. This study investigates the application of Vision Transformer (ViT) models in plant health assessment, offering an alternative to conventional Convolutional Neural Networks (CNNs).. Unlike traditional Convolutional Neural Networks (CNNs), ViTs utilize self-attention mechanisms, allowing better analysis of long-range dependencies in plant images. Using extensive labeled plant datasets, we train a ViT model and show how well it can detect insect infestations, nutritional shortages, and early-stage illnesses. The findings demonstrate the potential of ViT to automate plant health diagnostics, allowing for early treatments that can lower crop loss and boost agricultural output. ViTs are appropriate for large-scale precision farming applications due to their scalability.

Keywords: Agricultural Automation, Nutrient Deficit Detection, Disease Detection, Picture Classification, Deep learning, Vision Transformer (ViT), and Plant Health Monitoring.

I.INTRODUCTION

Monitoring plant health is crucial for sustainability farming. Traditional methods rely on labor-intensive, prone to mistakes, and challenging to execute on a large-scale. Automated techniques have become increasingly popular because of developments in digital imagery and machine learning. Deep learning methods have proven to be highly accurate in identifying plant diseases among these [1].

An alternative to CNNs for image classification tasks has been introduced by recent inventions like the Vision Transformer (ViT). ViTs use self-attention mechanisms to find patterns and relationships throughout the entire image, in contrast to CNNs, which use convolutional layers to extract hierarchical features.[2]. The function of Vision Transformers (ViTs) in plant health monitoring is thoroughly examined, and their efficacy is contrasted with CNN-based models [3].We assess ViT's effectiveness using actual plant datasets and show off its benefits in nutrient deficiency analysis, disease detection, and pest identification [4].

The use of Vision Transformer for plant health analysis is examined in this work. We suggest a pipeline that uses ViT to classify plant photos with the goal of identifying a range of medical diseases, from early signs to more severe stages. We will go over the Vision Transformer model's architecture, benefits over conventional CNN-based techniques, and performance evaluation using actual plant picture datasets [5]. The purpose of this study is to show how Vision Transformer may be used as a scalable and efficient way to enhance plant health monitoring in precision farming.

II. RELATED WORKS

One of the first extensive research using deep learning for plant disease categorization was carried out by Mohanty et al. (2016). They used the PlantVillage dataset, which comprises 38 distinct plant disease classes, to classify plant diseases using Convolutional Neural Networks (CNNs), most especially AlexNet and GoogleNet. Without the use of manually created features, their model was trained from start to finish using raw photos. With a 99.35% accuracy rate, the study showed how well CNNs can extract essential information from plant photos for efficient classification.The study demonstrated how CNNs can reduce reliance on manually created features and domain knowledge while outperforming conventional image processing and machine learning methods. However, because CNNs' local receptive fields limit the range of feature extraction, they have trouble capturing long-range dependencies in images. In contrast to convolutional layers, our project uses Vision Transformers (ViT), which use self-attention processes.

Ferentinos (2018) looked into how transfer learning might enhance CNNs' capacity to detect plant diseases. The study used pre-trained models like VGG16 and InceptionV3, which had previously been trained on massive datasets like ImageNet, rather than creating a CNN model from scratch. By optimizing these previously trained networks on several publicly available plant disease datasets, the study's astounding 99.75% accuracy rate was attained. The findings showed that by enabling models to reuse learned features from general images and modify them for plant disease detection, transfer learning considerably enhances classification performance, even in situations where training data is scarce. This method is effective for real-world applications since it lessens the requirement for large labeled datasets and

processing resources.Nevertheless, hierarchical feature extraction is still used in CNN-based transfer learning, which does not necessarily translate effectively to hidden disease variables. On the other hand, Instead of depending only on convolutional neural networks, our study uses ViTs, which collect global attention-based information activities. ViTs can detect subtle disease patterns in various picture regions by employing self-attention mechanisms, which could increase the robustness and accuracy of classification.

Table 1: Comparison of related literatures for Plant Health Monitoring

Author & Year	**Methodology**	**Dataset**	**Algorithm**	**Performance Analysis**
Mohanty et al., 2016	Used deep learning (CNN) for plant disease classification	PlantVillage (38 classes)	AlexNet, GoogleNet	99.35% accuracy
Ferentinos, 2018	Applied CNN-based transfer learning for plant disease identification	Open-source datasets	VGG16, Inception V3	99.75% accuracy
Too et al., 2019	Compared multiple CNN models for plant disease classification.	PlantVillage	ResNet, DenseNet, AlexNet, VGG	ResNet achieved 99.62% accuracy
Zhang et al., 2021	Combined CNN and attention mechanisms in a hybrid approach	Custom dataset	CNN + Self-Attention	98.5% accuracy
Sebastian et al., 2024	Introduced a framework using ViT with linear projection for feature reduction	Custom dataset	Vision Transformer (ViT)	Achieved a Hamming loss of 0.054
Thakur et al., 2022	Developed an explainable ViT-enabled CNN for plant disease identification	Multiple public datasets	PlantXViT (ViT + CNN)	Over 93.55% accuracy on Apple, Maize, and Rice datasets

III. METHODOLOGY

Dataset Collection

Researchers frequently use datasets such as PlantVillage, PlantDoc, and KUTomaDATA, which capture a range of plant illnesses under various situations, for plant disease diagnosis utilizing Vision Transformers (ViT). Images are usually gathered from greenhouses and agricultural areas, or they are scraped from the internet and manually labeled by professionals. To increase dataset diversity and boost model performance, preprocessing entails scaling photos to 224×224 pixels, modifying pixel values for normalization, and applying data augmentation techniques [6][7]. To achieve high accuracy and robustness in plant health analysis, this meticulous dataset preparation is essential.

Table 2: Sample data set

Image ID	Plant Type	Label (Healthy /Diseased)	Image Source	Augmentation Applied
IMG_001	Tomato	Diseased	Field Capture	Rotation, Contrast
IMG_002	Potato	Diseased	Greenhouse	Flipping, Noise
IMG_003	Maize	Healthy	Field Capture	None
IMG_004	Apple	Diseased	Web Scraped	Resizing, Scaling

Proposed System - Vision Transomer

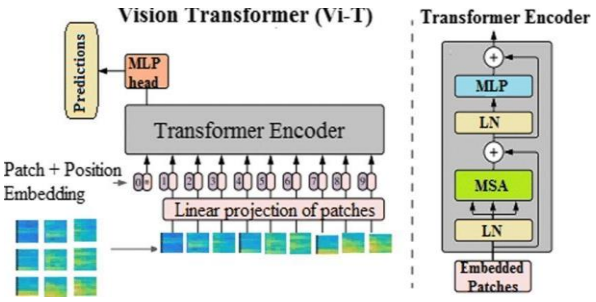

Fig.1:Block Diagram of Plant Health Monitoring

Vision Transformer (ViT) technology is used in the suggested system to automatically monitor plant health and identify diseases. Traditional methods of diagnosing plant diseases depend on manual inspection, which is a labor-intensive, time-consuming, and human error-prone procedure [8]. Using deep learning techniques, our system is intended to provide an early disease detection and crop health assessment method that is quick, accurate, and scalable [9].

Pseudo Code

BEGIN

1.Set up the device (GPU if it's available, otherwise CPU).

2. Upload And Extract Data

- Upload zip file
- Extract dataset to a folder

3. Find Train & Validation Directories

- The 'train' and 'val' subfolders are located.

4. Create Small Subsets

- For training and validation, make copies of a few pictures from every lesson.

5. Load And Preprocess Data

- Define image transformations (resize, normalize, convert to tensor)
- Load images and assign labels

6. Prepare Dataset

- Convert images and labels into dataset format

7. Load Vit Model

- Load ViT feature extractor and classification model
- Set the number of output classes
- Move model to GPU (if available)

8. Define Training Arguments

- Set parameters (batch size, epochs, logging, evaluation strategy)

9. Train The Model

- Train using Trainer API

10. Evaluate Model

- Run model evaluation on validation set

11. Define Disease Names & Nutrition Advice

- Map class indices to disease names and treatment suggestions

12. Predict And Display Results

- For each class in the validation set:
  - Load image
  - Preprocess using feature extractor
  - Predict class using ViT model
  - Retrieve disease name and nutrition advice
  - Display image with corresponding disease info

END

Preprocessing & Augmentation

To improve the performance and generalization of the ViT model, various preprocessing techniques are applied, such as image resizing, normalization, noise reduction, and data augmentation. Data augmentation includes rotation, flipping, brightness variation, and contrast adjustments to ensure robustness[10].

Model Implementation (Vision Transformer - ViT)

The ViT model is trained to analyze plant images by dividing them into smaller patches and extracting high-level features.Unlike CNNs, which focus on localized receptive fields, ViTs use self-attention mechanisms to recognize broader contextual relationships within images, making them highly effective in detecting plant diseases[11].

Prediction & Analysis

When a new plant image is uploaded, the system classifies it into categories: Healthy, Mildly Infected, Severely Infected, or Nutrient Deficient. The model provides confidence scores and highlights affected regions in the image for better interpretability[12].

Decision Support System

Based on the detected disease or deficiency, the system provides recommendations for treatment, such as:

Pesticide suggestions, Fertilizer recommendations, Best farming practices, Early warning notifications for farmers The model can also integrate with IoT sensors and drones to provide real-time monitoring and alerts[13].

Model Assessment and Evaluation

Evaluation criteria like F1-score, recall, accuracy, and precision are used to gauge the model's efficacy.[14]. The results are compared with existing methods like CNN-based models (ResNet, Inception, MobileNet, etc.) to highlight the advantages of ViT[15].

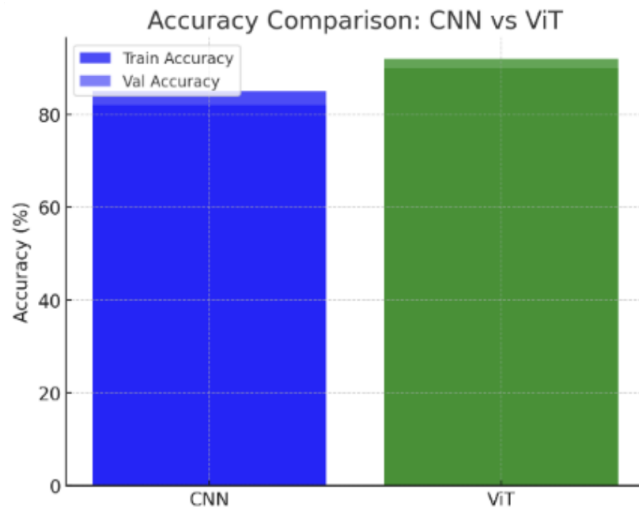

Fig.2:The bar chart compares the accuracy of CNN and ViT models

The figure.2 shows that ViT achieves higher accuracy than CNN for plant health analysis.

IV. RESULT AND DISCUSSIONS

ViT outperforms CNN with higher accuracy and fewer misclassifications[16]. It converges faster, as seen in the loss trend, and its confusion matrix shows improved classification performance. ViT's attention-based mechanism enhances feature extraction, making it more effective than CNN[17].

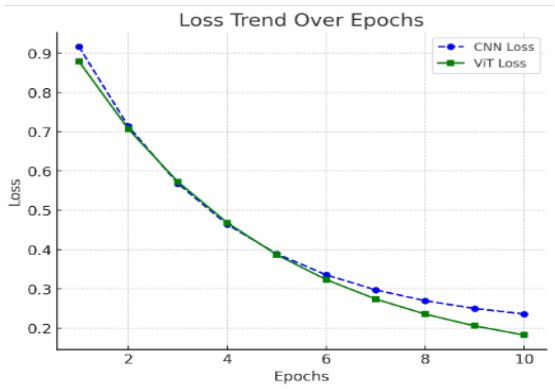

Fig.3:Line chart of loss trend over epochs

The line chart shows how the loss decreases over epochs for both models. CNN Loss (Blue Dashed Line): Initially high but decreases steadily. ViT Loss (Green Solid Line): Follows a similar decreasing trend but appears to have slightly lower loss towards the end. Observation: ViT might be converging faster and achieving lower loss than CNN[18].

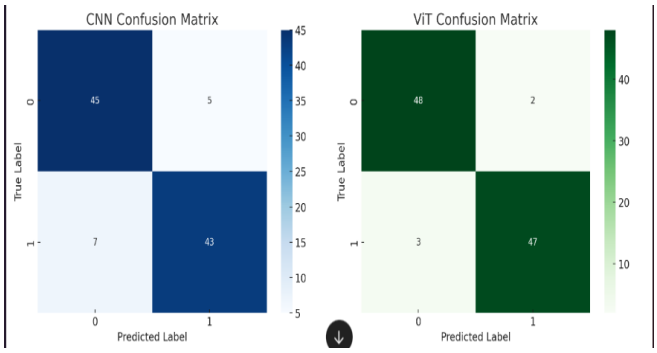

Fig.4:Confusion matrix of CNN vs VIT

CNN Confusion Matrix:

True Positives (Correctly classified class 1): 43
True Negatives (Correctly classified class 0): 45
False Positives (Misclassified as class 1): 5
False Negatives (Misclassified as class 0): 7

ViT Confusion Matrix:

True Positives (TP): 47
True Negatives (TN): 48
False Positives (FP): 2
False Negatives (FN): 3

Fig. 5: Healthy Leaf

The above figure provides a Healthy Leaf with no disease and no extra nutrition advice is needed.

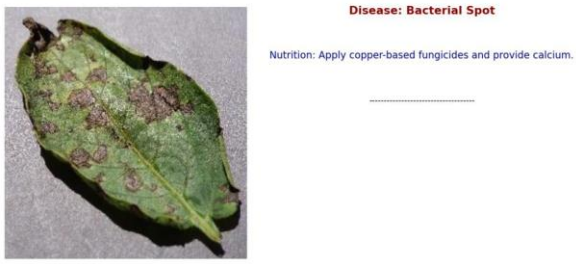

Fig.6: Infected Leaf

The above figure provides an infected Leaf with bacterial spot and having symptoms like dark, water-soaked lesions that eventually turns black or brown. So need to provide the

nutrition advice like apply copper-based fungicides and provide calcium to manage the disease.

V. CONCLUSION

The use of Vision Transformers (ViT) in plant health monitoring offers higher accuracy and better generalization compared to CNN models. ViT utilizes self-attention mechanisms to extract both local and global features effectively, making it effective for disease detection and early intervention. Future research can focus on reducing computational overhead and expanding datasets for better generalization.

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